

Please select one scenario among these two for your software engineering module project.

Scenario 1

A busy mid-sized restaurant needs a new management system to address several key requirements. The restaurant has been using outdated methods and is facing issues with efficiency and accuracy. The new system should simplify and improve various aspects of restaurant management.

Firstly, the system must handle reservations efficiently. Customers should be able to book tables online, view real-time availability, and receive confirmation and reminders. Each reservation will be tracked with attributes such as Reservation ID, Customer Name, Table Number, Date and Time, Number of Guests, and Status. Restaurant Staff will need a clear, organized view of all upcoming reservations to manage seating and customer flow effectively.

Secondly, the system needs to streamline order management. Waitstaff should be able to input orders directly into the system from tablets or terminals, which will include attributes such as Order ID, Table Number, Items Ordered, Quantity, and Special Requests. Kitchen Staff will receive these orders in real time, and the system should track order progress, providing updates on the status of each order to notify waitstaff when orders are ready.

Another critical requirement is inventory management. The system should monitor stock levels in real time, with attributes including Item ID, Item Name, Quantity on Hand, Reorder Level, and Supplier Information. It should alert Inventory Staff when items are running low and facilitate easy reordering. The system should also provide reports on inventory usage to help with planning and reducing waste.

The system must also support staff scheduling. Managers should be able to create and manage staff schedules, with attributes such as Staff ID, Name, Position, Shift Times, and Hours Worked. The system should handle time-off requests and allow Staff Members to view their schedules, request changes, and check their hours.

Additionally, the system should integrate with existing hardware, such as point-of-sale systems, and be secure to protect sensitive customer and business data. It should be user-friendly and accessible, requiring minimal training for Staff to use effectively.

Finally, the system should be reliable, with minimal downtime and strong Customer Support available to address any issues that arise. By addressing these requirements, the new management system will help the restaurant improve its operational efficiency, reduce errors, and enhance the overall dining experience for Customers.

Scenario 2

A large game development studio is tackling the challenge of employee fatigue, particularly during high-stress periods like "crunch time," by developing an **Employee Fatigue Monitoring Game**. This interactive game will assess and monitor employee fatigue levels through a series of cognitive and physical tasks designed to measure factors like focus, reaction times, and mental agility. The system will be integrated into the daily routine of employees, serving both as a fun and engaging tool and as a serious means to track and prevent burnout. By leveraging gamification and modern software engineering principles, the system will be reliable, secure, scalable, and adaptable to the unique needs of the studio's workforce.

The game will feature a variety of **cognitive and reaction-based tasks** that test different cognitive and motor functions. These include **Reaction Time Challenges**, where employees complete quick-reflex tasks like clicking on fast-moving objects or responding to on-screen cues within a limited time. Declining reaction times could indicate mental fatigue. Additionally, **Memory Tests** will challenge employees to remember sequences, patterns, or visual cues over short periods. As memory performance diminishes, this could point to signs of tiredness or overwork. In **Focus and Attention Tasks**, employees will be required to complete activities involving sustained attention, such as spotting differences between images or completing simple puzzles under time pressure. Errors or a decline in focus during these tasks will be flagged as indicators of fatigue. Lastly, **Decision-Making Scenarios** will test the ability to make quick choices under pressure, with slower or poor decision-making outcomes contributing to the overall fatigue score.

The system will automatically gather **integrated performance metrics** from each game session, tracking attributes such as **Reaction Time**, **Accuracy**, **Errors Made**, and **Completion Time** for each task. As employees play over time, the game will build a baseline of their performance, detecting trends or declines in mental acuity, concentration, and reflexes. The data will then be used to calculate a **Fatigue Score**, which can be compared against thresholds to determine if the employee is showing signs of fatigue.

One of the game's standout features is **dynamic difficulty adjustment**, which adapts to the player's performance and biometric data. If the system detects that the employee is showing signs of fatigue, such as slower reaction times or increased errors, the tasks may become slightly easier, or the game might prompt the employee to take a break. This encourages real-time awareness of fatigue while maintaining a positive and engaging experience.

In addition to gameplay data, the system will feature **daily check-ins and surveys**. These quick self-assessment surveys, completed before or after game sessions, will ask questions about how rested employees feel, their mood, and any physical discomfort. The subjective data will be cross-referenced with performance metrics and biometric data to offer a holistic view of each employee's fatigue levels.

After each session, employees will receive a **Fatigue Score**, calculated from game performance and self-reported feedback. The system will track scores over time, providing visual feedback such as fatigue trends, daily performance variations, and recommendations for well-being improvement, such as taking breaks or adjusting work hours. This score will also be made available in aggregate form to managers and HR teams, who will receive detailed reports on team-wide fatigue levels without compromising individual privacy.

Based on the employee's fatigue levels, the game will offer **personalized recommendations**. These could include encouraging the employee to take a short break if signs of fatigue are detected, offering lifestyle suggestions to improve sleep hygiene or manage stress, or identifying patterns where prolonged overwork is leading to deteriorating performance. This data will help both employees and managers adjust schedules or project timelines accordingly.

A key feature for management is the **dashboard interface**, which aggregates data across teams to identify trends in employee fatigue. The dashboard will provide insights into overall team well-being, alerting managers to potential issues before they become critical. HR and managers can filter data by department, role, or time periods, enabling them to make informed decisions, such as adjusting workloads, rescheduling tasks, or offering additional rest periods during high-stress periods.

By incorporating these features, the **Employee Fatigue Monitoring Game** will provide the game development studio with a novel and engaging method to track and identify fatigue in employees. This system will help prevent burnout and overwork, fostering a healthier and more sustainable work culture while utilizing the studio's expertise in game development.